K810 — CP treatment (workstream B), with a self-correction. Parity and CP are NOT "one read-direction, two shadows" — that was an over-unification. They are TWO DIFFERENT orientations: parity = the SPINOR chirality (discrete, real, from n_C odd → maximal); CP = the COMPLEX-PHASE orientation (continuous, from the S^1 → free). CP is CP-even under chirality, so it's a separate structure. The discrete-vs-continuous character is exactly what explains maximal-P vs free-CP — and that's the real, defensible content.

Keeper | 2026-07-22 | Working the CP workstream. Discipline fired on my own framing: "one ω, two shadows" (what I told Casey) doesn't survive scrutiny. Correcting it, and the corrected picture is tighter + consistent with everything banked.

★ Self-correction (the discipline firing on me)

I pitched parity and CP as "two shadows of one read-direction ω." Wrong on examination. Parity acts on the spinor chirality (real, discrete Z_2); CP acts on the complex phase (continuous, S^1). And crucially: CP is CP-EVEN under chirality — CP maps a left-handed chiral coupling back to a left-handed chiral coupling, so the *chirality* (parity's structure) is *preserved* by CP. CP violation is the *phase* residual, a different object entirely. So they are two different orientations of two different structures, not one ω. Retract the "one object" claim.

The corrected, honest unification (this is the content)

The geometry carries two kinds of orientation, and their *character* follows from their *type*: - Parity = the SPINOR orientation — real, DISCRETE (Z_2, $\omega=\pm1$ from n_C odd). A discrete orientation, gauged by a group projector → MAXIMAL (100%, V−A). (Derived-conditional, K809.) - CP = the COMPLEX-PHASE orientation — the phase on the S^1 of the Shilov boundary, CONTINUOUS. A continuous angle the radial geometry can't fix → FREE (= our banked δ_PMNS-free, K791/K792). Discrete → maximal; continuous → free. That is the reason parity is maximal but CP is free — not one object, but two orientations whose *type* (discrete vs continuous) sets their character.

The P / C / CP pattern, explained from this

- P maximal, C maximal: both from the SPINOR chirality (n_C odd) — the weak current is purely left, so P and C are each maximally violated. Same discrete source.
- CP small/free: because CP *preserves* the chirality (CP-even), the discrete maximal P- and C-violations do NOT show up in CP — they map weak-states to weak-states. The ONLY residual that can violate CP is the continuous complex phase. So CP ≈ conserved, violated only by the free phase. This is why P, C are maximal but CP is tiny — the discrete violations are chirality, which CP respects.
- CP possible □ 3 generations = rank+1: with 2 generations the phase is removable (0 physical phases); 3 gives exactly 1 physical phase. So CP violation *can exist* precisely because there are 3 generations = rank+1 (rank=2). Structural link.
- Magnitude of CKM CP small: not from the phase (free) but from the small quark mixing angles — rank-1 suppression, banked K788 (CP small = rank-1). J = (free phase) × (small rank-1 angles).

The T / arrow-of-time connection — real but bounded (flag, do NOT bank)

CPT is a theorem → CP violation □ microscopic T-violation (a coupling phase). That's solid. BUT microscopic T ≠ the thermodynamic arrow of time — do not conflate. There IS a genuine downstream link (CP violation is a Sakharov condition for baryogenesis → matter dominance → tied to initial conditions/the arrow), but it's subtle and multi-step. So: CP ▯ microscopic-T (solid); CP → baryogenesis → matter (real, subtle); CP = “the arrow of time” (over-reach — resist). I-tier lead at most.

Honest tier + consistency check

- I-tier organizing principle (a *reason* for the P/C/CP pattern), corrected from the over-unification. NOT new physics.
- Consistent with everything banked: parity maximal (K809, from n_C odd) ▯; δ_PMNS free (K791/K792, the continuous-phase statement) ▯; J_CKM small = rank-1 (K788) ▯; CP small because chirality is CP-even ▯. It doesn’t add a new number; it *explains the pattern* from the two-orientations structure.
- What it does NOT do: predict the value of δ (free) or J (rank-1 angles × free phase). No new prediction — a reason, per Casey’s frame.

For Elie / Grace (verify)

- ELIE: confirm the counting — physical CP phase exists ▯ ≥ 3 generations (rank+1); J_CKM = phase × angles with the angles rank-1-suppressed; the “CP-even chirality” statement (CP preserves the left-handed current).
- GRACE: render the TWO-orientations map — spinor(discrete→parity maximal) vs phase(continuous→CP free) — NOT one ω; keep the self-correction visible so we don’t re-merge them.

— Keeper K810, 2026-07-22. SELF-CORRECTION: parity and CP are NOT one read-direction — parity = spinor chirality (discrete, real, n_C-odd → maximal); CP = complex-phase orientation (continuous, S^1 → free). CP is CP-even under chirality, so the discrete P/C violations cancel in CP, leaving only the free continuous phase → CP small. Discrete→maximal, continuous→free explains why P maximal / CP free. CP possible ▯ 3 gens=rank+1; J small from rank-1 angles (K788). CP▯micro-T (CPT); ≠ thermodynamic arrow (flag). I-tier, corrected, consistent with all banked, reasons-for-SM not new physics. See [[Keeper_K809_info_theory_code_thread_progress_parity_from_nC_odd_message_syndrome_partition_2026-07-22]], K791/K792 (δ free), K788 (CP small=rank-1).